

A.4 : Hall-Effect Sensor RPM Measurements compared to Tachometer Readings

Voltage	Tachometer Readings (RPM)				Arduino Calculated Value (RPM)				Absolute Error	Percentage Error
	1	2	3	Average	1	2	3	Average		
1	131.2	131.4	131.3	131.3	132	132	131	132	-0.7	0.54
2	350.2	350.4	350.5	350.3	350	351	350	350	0.3	0.10
3	565.9	565.9	566.1	566.0	566	566	566	566	0.0	0.00
4	759.1	759.2	759.3	759.2	759	760	759	759	0.2	0.03
5	982.8	983.0	982.9	982.9	983	983	983	983	-0.1	0.01
6	1175	1175	1175	1175	1175	1175	1175	1175	0.0	0.00

A.5 : Hall-Effect Sensor Working Distance (mm)

1	2	3	Average
3.06	3.00	3.05	3.04

A.6 : IR Optical Sensor RPM Measurements compared to Tachometer Readings

Voltage	Tachometer Readings (RPM)				Arduino Calculated Value (RPM)				Absolute Error	Percentage Error
	1	2	3	Average	1	2	3	Average		
1	128.5	128.6	128.8	128.6	129	128	130	129	-0.4	0.29
2	352.5	352.7	352.7	352.6	350	351	350	350	2.6	0.75
3	565	565.1	565.2	565.1	566	567	567	567	-1.9	0.34
4	760.1	760.4	760.2	760.2	760	760	760	760	0.2	0.03
5	982.8	982.8	983.0	982.9	983	984	983	983	-0.1	0.01
6	1174	1174	1174	1174	1174	1174	1174	1174	0.0	0.00

Table A.7 : Optical Sensor Working Distance (mm)

1	2	3	Average
12.95	12.98	13.01	12.98